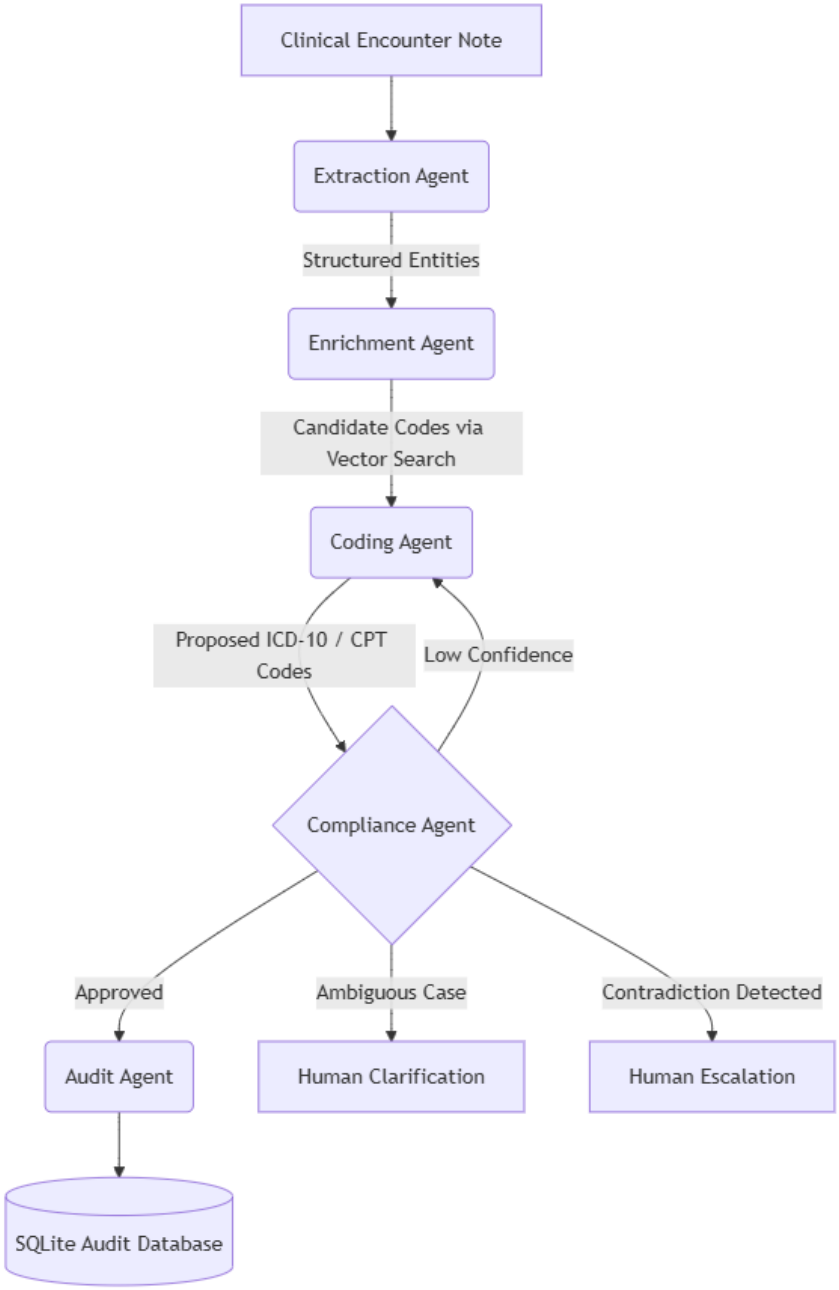
PARS AI Agent

Multi-Agent Architecture for Autonomous Medical Coding & Compliance

1. System Overview

This system is an **enterprise-grade multi-agent AI architecture** designed to automatically process clinical encounter notes, retrieve relevant ICD-10/CPT codes using semantic retrieval, apply clinical reasoning to assign codes, enforce payer-policy compliance, and log decisions for auditability and operational metrics.

2. High-Level Architecture



The pipeline ensures that **coding decisions pass through compliance guardrails before final approval**, minimizing billing errors and ensuring payer-policy adherence.

3. Agent Roles

Extraction Agent

Purpose

Convert unstructured clinical notes into structured medical entities.

Model

llama-3.1-8b-instant

This lightweight model is used for **fast, cost-efficient NLP extraction tasks**.

Enrichment Agent

Purpose

Retrieve possible medical codes related to extracted entities using vector search.

Tools

- ChromaDB
- Sentence-transformers embeddings (all-MiniLM-L6-v2)

Coding Agent

Purpose

Apply deep clinical reasoning to determine appropriate codes.

Model

llama-3.3-70b-versatile

This model is used only for **complex clinical reasoning tasks**.

Compliance Agent

Purpose

Ensure coding decisions follow payer policies and regulatory rules.

Possible Outcomes

- Approved
- Reflection required
- Clarification required
- Escalation required

Audit Agent

Purpose

Provide system traceability and operational metrics.

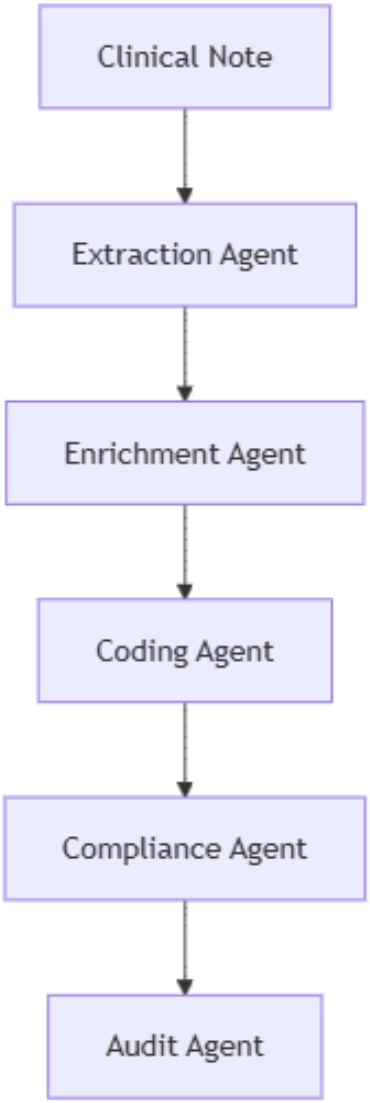
Storage

SQLite Database

4. Agent Communication Model

Agents communicate through a **shared LangGraph state object**.

Example Interaction Flow



5. Error Handling and Guardrails

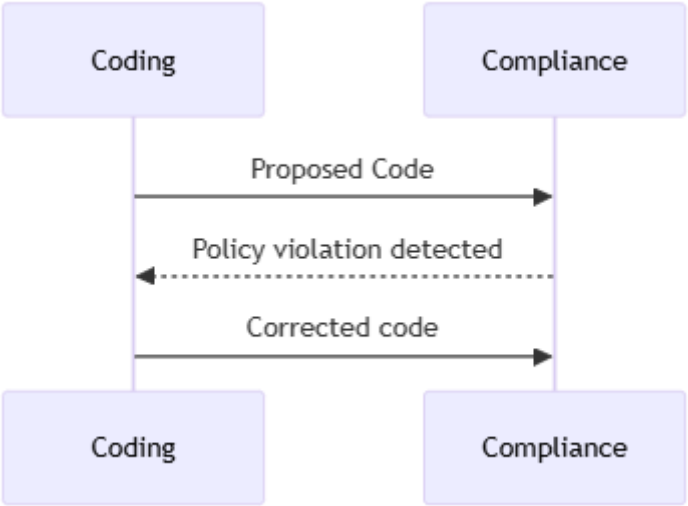
The architecture includes several **robust safety mechanisms**.

Self-Reflection Loop

If the Compliance Agent detects a policy violation or hallucinated code, it triggers a reflection loop.

Compliance → needs_reflection = True

The pipeline automatically routes back to the Coding Agent with feedback.



Escalation Logic

Certain scenarios require manual review.

Examples include:

- Contradictory clinical information
- Missing documentation
- Confidence score below threshold

Graceful Degradation

If LLM services become unavailable, the system falls back to vector retrieval.

Fallback pipeline:

Extraction → Vector Retrieval → Best Matching Code

This ensures **continuous operation in clinical environments**.

6. Observability and Metrics

All decisions are logged in the audit database.

Tracked metrics include:

- Autonomous coding success rate
- Escalation frequency
- Average confidence score
- Throughput performance

Example savings calculation:

Savings = Autonomous Cases × $35 manual coding cost

These metrics power the **React performance dashboard** used for operational insights.

Summary

The Healthcare Operations AI Agent architecture combines:

- Multi-agent AI orchestration
- Retrieval-Augmented Generation
- Compliance rule engines
- Human-in-the-loop escalation
- Full audit traceability

This modular design enables **scalable, compliant, and autonomous clinical coding workflows** while maintaining strict operational safeguards.